

Hersh Patel

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EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN (ONLINE)

Masters of Science

Major: Computer Science

Austin, TX

Expected May 2028

RUTGERS UNIVERSITY (GPA: 4.0)

Bachelor of Science

Major: Computer Science

New Brunswick, NJ

May 2026

TECHNICAL SKILLS

Languages and Frameworks: Python, Java, C#/.NET, TypeScript, JavaScript, HTML/CSS, SQL, MATLAB, Node.js, Angular, React, .NET Core, Entity Framework Core, FastAPI, pandas, NumPy, Matplotlib, seaborn, psycopg, JDBC

Cloud, Databases & DevOp: AWS (S3, DynamoDB, Lambda, EC2, IAM, CloudFormation, AWS SDK), Azure (PostgreSQL, Cognitive Services), Git, GitLab CI/CD, MSSQL Server 2019, PostgreSQL, MySQL, MongoDB, Supabase, Firebase

AI / ML & Tooling: PyTorch, scikit-learn, LangChain, FAISS, OpenAI Embeddings API

Certification: AWS Certified Developer – Associate, Azure Developer Associate

CLASSES

Machine Learning Principles: supervised/unsupervised learning, model evaluation, bias-variance tradeoff, **regularization**, ensemble methods

Intro to Data Science: data wrangling, EDA, statistical inference, **visualization**, **end-to-end ML pipelines**

Data Structures: arrays, linked lists, trees, graphs, hash maps, sorting/searching algorithms, time/space complexity

Discrete Structures: combinatorics, graph theory, propositional logic, proof techniques, recurrence relations

Linear Algebra: matrix decomposition, eigenvalues/eigenvectors, vector spaces, SVD, transformations

Linear Optimization: LP formulation, **simplex method**, duality theory, **convex optimization**, **constraint modeling**

Data Management for Data Science: SQL, relational schemas, query optimization, NoSQL, **data pipelines**, ETL

Design & Analysis of Computer Algorithms: **dynamic programming**, **greedy algorithms**, NP-completeness, divide & conquer, graph algorithms, complexity analysis

Principles of Information & Data Management: **database design**, normalization, indexing, transaction management, ACID compliance, **data integrity**

EXPERIENCE

Charles Schwab

Software Development Engineer

Austin, TX

Jun 2026 – Present

- Contributed to the full Software Development Lifecycle by collaborating cross-functionally to modernize a Retirement Plan Services platform through a **Domain-Driven Design** migration, improving system scalability and long-term maintainability while delivering features in an **agile team environment** using .NET Core and C#.
- Improved developer experience** and **reduced integration** effort for consuming teams by designing and implementing full end-to-end RESTful API endpoints and authoring OpenAPI specification documentation establishing a single source of truth for all endpoint contracts, request/response schemas, and error definitions backed by optimized MSSQL queries and object-oriented design principles.
- Enhanced system reliability and reduced mean time to resolution for production incidents by implementing structured observability pipelines using **Splunk**, while integrating AI-assisted development tools to eliminate manual bottlenecks, increase engineering throughput, and deliver higher-quality solutions that support client-facing financial services.

MSH Global

AI Engineer Intern

Hillsborough, NJ

Jun 2025 – Aug 2025

- Architected an **AI-powered policy chatbot** for insurance employees using **LangChain**, **PostgreSQL on Azure**, and a **FAISS vector store** with **OpenAI Embeddings** to perform semantic search across thousands of billing and policy documents, cutting average query **latency by 50%** and eliminating manual document lookups across teams.
- Integrated **SQLDatabaseChain** with a **LangChain SQL agent** to automate structured insurance query resolution against a live relational database such as standardizing query logic, reducing human error, and cutting data retrieval time by over 75% for high-frequency operational workflows.
- Designed and deployed a customer-facing **Node.js UI** on top of the chatbot backend, abstracting the underlying LLM and database infrastructure so non-technical insurance companies could independently configure and operate the tool without engineering support.

Snowlake Solutions

Full Stack Software Developer Intern

Austin, TX

Jun 2024 – Aug 2024

- Developed a production **.NET Core C# REST API** for a Portfolio Management System supporting real-time client account and holdings search, using Generics, LINQ, async/await, and **Entity Framework Core**, and shipped to production via a fully automated **CI/CD pipeline on GitLab**.
- Authored stored procedures, scalar functions, and indexed views in MSSQL Server 2019 to optimize complex multi-table data operations across client account datasets, ensuring transactional consistency and testability across environments.
- Built interactive **Angular 17 (TypeScript)** UI components with real-time data binding, enabling employees to query, filter, and visualize portfolio holdings with significantly improved response times over the legacy interface.

PROJECTS

NBA Score Predictor | *Python, PyTorch, scikit-learn, NumPy, pandas, matplotlib, seaborn* | Dec 2025

- Engineered an end-to-end ML prediction pipeline with modular *scikit-learn* feature transformation, custom *PyTorch Neural Network* (stacked Linear layers, ReLU, Dropout, BatchNorm), and Adam optimizer with LR scheduling and Huber Loss, achieving MAE < 5 pts and MAPE < 5.5% on held-out test data, outperforming linear regression and ensemble baselines across all CV splits
- Conducted rigorous EDA via matplotlib/seaborn, KDEs, residual plots, correlation heatmaps, and distributional drift analysis, driving feature selection and dimensionality reduction decisions that directly improved convergence speed and generalization by 30%

Account Documents Migration System | *.NET Core, C#, AWS S3, DynamoDB, IAM, AWS SDK* | Sep – Nov 2025

- Architected a production-grade cloud-native .NET Core microservice leveraging AWS SDK to automate **large-scale migration** of thousands of legacy documents to S3 via a fault-tolerant, event-driven ingestion pipeline with zero-trust IAM least-privilege access and regex-based document classification for automated S3 folder routing
- Designed a **master-worker concurrency architecture** with dynamic thread pool autoscaling, async/await parallelism, and DynamoDB-persisted audit logs with conditional writes and optimistic locking for idempotent deduplication and point-in-time rollback

Raillex | *Java, JSP, MySQL, JDBC, HTML/CSS, JavaScript* | Jun – Jul 2025

- Architected a full-stack multi-tier train reservation platform using Java Servlets, JSP, and MVC, implementing RBAC with session management and **auth middleware**, backed by a 3NF-normalized MySQL schema with composite indexes and ACID-compliant parameterized JDBC queries with connection pooling to prevent SQL injection under high-concurrency load
- Shipped a responsive UI with CSS Flexbox/Grid and vanilla JS, featuring **client-side validation**, **dynamic DOM rendering**, and async data fetching across customer and employee portals